The cognitive turn in experimental pragmatics: Towards a computational cognitive science of communication

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Abstract

concrete.

1 Experimental pragmatics and computational modeling

Pragmatics studies how language is used for efficient communication and how communicated meaning is modulated, if not created, in a particular context by systematic, context-general patterns of use. Its origins lie in the philosophy of language, in particular the analytical tradition of philosophy, in work on topics like *speech-acts* (Austin, 1962; Searle, 1969), *conversational implicature* (Grice, 1975), or *presupposition* (Stalnaker, 1973; Karttunen, 1973). Much early work on these topics remained invested in a logical-formal approach (Gazdar, 1979; Kadmon, 2001), but the early 2000's brought a deep and empowering incision: a widely adopted and ultimately very successful *experimental turn* (Noveck and Sperber, 2004; Noveck, 2018).[1]

The experimental turn in pragmatics was accompanied by formal and computational work from the start, with two main strands of approaches emerging at opposite ends of a spectrum. On one end of the spectrum, *grammaticalism* focuses on language as an abstract, computationally encapsulated system [MF: refs]. Grammaticalism strives to give formally precise explanations for how pragmatic inferences (focusing almost exclusively on interpretation) could arise in a language-specific computational module [MF: refs]. On the other end of the spectrum, *probabilistic pragmatics* seeks explanations of pragmatic phenomena from the point of view that linguistic communication is a special form of social interaction, also taking aspects of domain-general cognition of language users into account [MF: refs]. Figure 1 gives a coarse-grained overview of the development of, in particular, probabilistic pragmatics as the culmination point of research that combines aspects of individual cognition, such as bounded rationality [MF: refs], and functional explanations based on optimal information transfer in communication [MF: refs].

While experimental data can mitigate between alternative explanations proposed in the grammaticalist tradition [MF: refs], I argue here that the more socio-cognitive approach adopted by probabilistic pragmatics is, if done right, much better suited to incorporate *both* linguistic theory

[1] This development paralleled similar empirical re-orientations in other hitherto very analytically minded subfields of linguistics like formal syntax (Sprouse, 2007, 2023) and formal semantics (Meibauer and Steinbach, 2011; Cummins and Katsos, 2019).

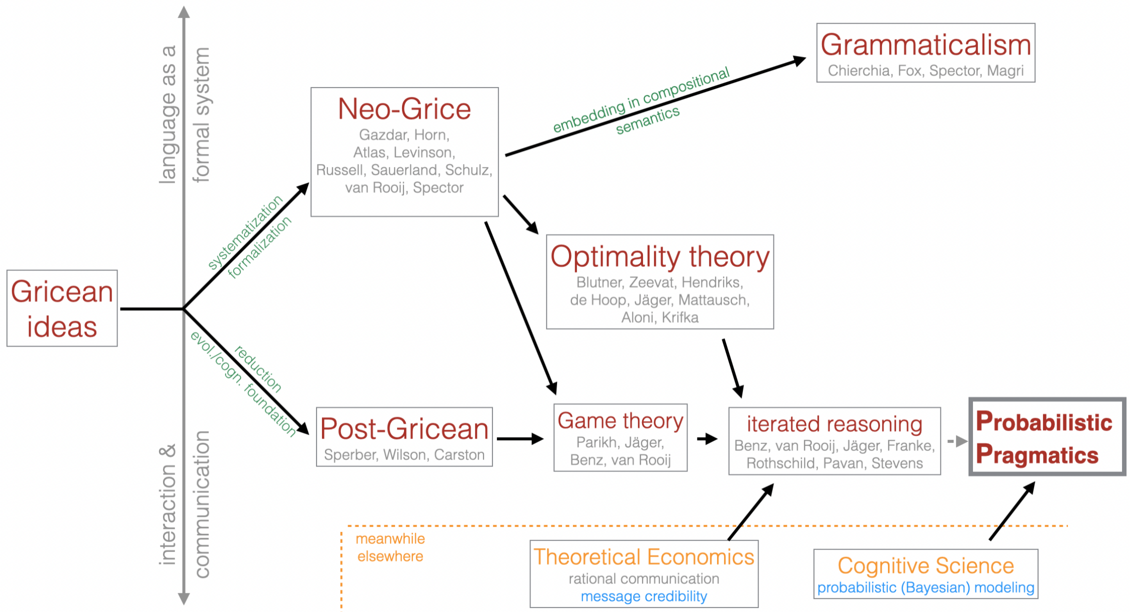

Figure 1: Genealogy of probabilistic pragmatics. [MF: describe what's pictured here (if you keep it; maybe make this figure and the text part of an InfoBox?)]

and aspects of general cognition (such as attention, meta-cognition, or memory), as well as the concrete affordances of a conversational situation or an experimental task, into a unified computational model of the *data-generating process*, i.e., the (cognitive) processes by which individual data observations, such as concrete data points recorded from an experiment, come about. Seen in this light, computational modeling in pragmatics can serve a bridging function between language, linguistic cognition and domain-general aspects of information-processing and decision-making. The central argument of this paper is, consequently, that the momentum gained from the experimental turn should be channeled into a dedicated *cognitive turn*, which seeks to explain language use as the intricate interplay between language-specific and domain-general cognition. Computational cognitive-pragmatic models play a pivotal role in this cognitive turn, as they require detailed commitment about the role of linguistic theory (latent, established concepts) in the data-generating process, and the way that general cognitive factors shape linguistic behavior. As I will argue here, recent work in modelling-oriented experimental pragmatics has started to "go cognitive" in this sense, but that a more dedicated embrace of the cognitive turn is needed. Looking forward, I sketch possible future directions.

[MF: spell out paper overview] Section [MF: XYZ] introduces probabilistic pragmatics models. Section [MF: XYZ] elaborates on the idea of cognitive models acting as theory-driven statistical models for the data-generating process. It covers some main lessons learned from probabilistic pragmatic modeling of experimental data, and points to key open questions for the future. Section [MF: XYZ] then focuses on a high-level case study, namely probabilistic reasoning about common ground, as an illustrative example …

2 Probabilistic pragmatics

Notable early probabilistic approaches to pragmatics include pioneering work by Merin (1994, 1999), Zeevat (2014), and Russell (2012). A particularly prominent style of probabilistic modeling in pragmatics is the Rational Speech Act framework (Frank and Goodman, 2012; Scontras, Tessler, and Franke, 2021; Degen, 2023), the basic set-up of which was anticipated in decision-theoretic work by Anton Benz (Benz, 2006; Benz and van Rooij, 2007). Early contributions in the evolving RSA tradition looked at implicature (Bergen, Levy, and Goodman, 2012; Goodman and Stuhlmüller, 2013), stylized reference games (Frank and Goodman, 2012; Degen and Franke, 2012; Degen, Franke, and Jäger, 2013) , or the place of probabilistic notions in semantics and pragmatics in general (Goodman and Lassiter, 2014).

What unites probabilistic approaches to pragmatics is a focus on explaining pragmatic phenomena in terms of what agents *do* in linguistic communication. More concretely, following the general tenets of Gricean Grice (1975) and Post-Gricean pragmatics (Sperber and Wilson, 1995; Wilson and Sperber, 2004; Blutner and Zeevat, 2004), an emergent view in concurrent cognitively oriented pragmatic modeling is that linguistic behavior is to be explained as REAL behavior (**r**esource-**e**fficient, **a**pproximately optima**l**). The assumption of (approximate) optimality of behavior allows for functional explanations, focusing on the *why* a particular behavior or general phenomenon is likely to be supported as a *good* solution to a recurring communication problem (e.g., Nowak and Krakauer, 1999; Skyrms, 2010). Resource-efficiency stresses that realizations of implementations of (approximately) optimal solutions may trade-in some ulterior success for easier representation or policy execution (e.g., Lieder and Griffiths, 2019; Lai and Gershman, 2024).

The assumption that linguistic communication is based on **REAL behavior** leads to naturally to a picture of pragmatic interpretation as a form of inverse, or abductive, reasoning (Hobbs, Stickel, and Martin, 1993; Bunt and Black, 2000; Franke and Jäger, 2016): since it is natural to assume that a speaker's linguistic behavior (their choice of words, the fact that they spoke in the first place, their intonation, or speech-accompanying gesture, etc.) is at least partly determined by a latent, i.e., unobservable, mental state of the speaker, the main task of pragmatic interpretation is to reason backwards, as it were, about how relatively likely or unlikely different conceivable causes of the speaker's behavior are.

In keeping with the idea of (approximate) optimality of REAL behavior, this type of abductive reasoning has a normatively compelling solution in Bayesian probabilistic inference. Given a probabilistic speaker policy $\pi(a \mid t)$, which specifies how likely a speaker is to perform an action $a \in A$ when in state $t \in T$, together with a prior distribution over how likely different states t are, the normatively correct way of updating these prior beliefs is given by Bayes' rule as follows:

$$P(t \mid a) = \frac{\pi(a \mid t)\, P(t)}{\sum_{t' \in T} \pi(a \mid t')\, P(t')} \tag{1}$$

To add content to this basic—and, at least as a starting point, arguably natural—picture of interpretation, at least three things need to be specified: (i) the relevant set of states T; (ii) the relevant set of actions A; and (iii) the speaker's behavioral policy π. Consider, for instance, a standard case of referential communication. Here, the states T could be taken to be the potential referents in the shared perceptual context between speaker and listener, while the actions A could be a finite set of utterances or descriptions, identified, say, at the level of surface form, such as "the blue triangle." The speaker's policy π could then be defined on the basis of standard assumptions about how speakers are expected to behave in situations of the relevant kind, for example, as envisaged by Grice's Maxims of Conversation (Grice, 1975). In particular, for a simple case of referential communication, we might define a probabilistic policy in terms of a utility function that scores each action (utterance) a in a given state t based on a's truth-value in t (quality), the degree to which a conveys information about t as opposed to any other $t' \in T$ (informativity), and the cost $C(a)$ of performing a:

$$\mathrm{U}(a, t) = \mathrm{Informativity}(a) + \mathrm{Truth}(a, t) + \mathrm{Cost}(a) \tag{2}$$

The speaker's policy is then defined as (approximately) optimal behavior for this utility function, for instance as implemented via the softmax function (Franke and Degen, 2023):

$$\pi(a \mid t) = \frac{\exp \alpha \; U(a, t)}{\sum_{a' \in A} \exp \alpha \; U(a', t)} \tag{3}$$

The standard formulation of the RSA speaker policy, as given by Frank and Goodman (2012) and much subsequent work, conforms to this general utility-optimization picture, but is usually stated in terms of a speaker optimizing their linguistic expressions relative to a literal listener. Since this literal listener is typically construed as a context-blind, unstrategic dummy, however, it is more useful to think of a probabilistic pragmatic model, more generally, as consisting of two components:

- a speaker policy that seeks to approximate optimal behavior under some (numeric) utility defining "efficient" or "successful" communication; and
- a Bayesian listener, who inverts this speaker policy in order to infer likely explanations for the observed linguistic behavior.

Many interesting variations, extensions and applications of this general approach to probabilistic pragmatic modeling have been developed over the years. These often extend the factors that feed into the speaker's utility function, e.g., to include aspects such as politeness or social meaning (e.g. Burnett, 2019; Yoon et al., 2020; Carcassi and Franke, 2023), or other aspects such as argumentativity (Carcassi, Wang, et al., 2026). [MF: insert more references] Another common extension is the inclusion of multiple latent variables for the listener to infer, each of which may shape the speaker's policy in different functional ways. Examples are the inclusion of a question under discussion (Kao et al., 2014), comparison classes [MF: refs], and more [MF: insert more examples]. All of these mentioned works and many others tightly link up with experimental data, either for testing categorical hypotheses supported by particular models, or, more frequently, to assess the quantitative fit of one or several models. If done right, this can be one of the assets that probabilistic modeling approaches bring to experimental pragmatics, as argued more extensively in the next Section 3.

Some models and applications also explore, or even require, the higher-order iteration of speaker policies and listener interpretations (e.g. Franke and Jäger, 2014; Bergen, Levy, and Goodman, 2016; Frank, Emilsson, et al., 2016). The general idea is that a speaker may optimize their linguistic behavior to be informative (or otherwise optimized) for a specific listener interpretation behavior that itself is optimized to the speaker's production policy. It then becomes an empirical question how much social reasoning is routinely employed by language users in general, or how much iterated perspective-taking individual language users are capable of [MF: refs]. However, from a resource-rational perspective on efficient cognitive processing (Lieder and Griffiths, 2019), we should expect to see meta-cognitive allocation of reasoning effort to be flexibly employed on-demand — a perspective that will be centrally spelled out in Section 4.

3 Theory-driven probabilistic modeling

A now-familiar starting point for any discussion of formal modeling is Box's dictum that all models are wrong, but some are useful (Box, 1979). Yet acknowledging that no model can be strictly true does not mean that all models are wrong to the same degree. Following Isaac Asimov's (1988) observation that wrongness itself admits of degrees—that the belief the Earth is a sphere is less wrong than the belief that it is flat, even though neither is exactly correct—we should prefer models that are less wrong over those that are more so. This raises the natural question of what makes such (less) wrong models useful in the first place.

Just as the familiar principle "garbage in, garbage out" governs the relationship between data quality and the conclusions we can draw from it, an analogous principle governs models themselves: the more (reasonable) assumptions we build in, the more we can conclude from the data. Causal modeling offers a clear illustration of this logic—by committing to a set of reasonable structural assumptions about how variables relate to one another, we are able to extract substantive, interesting conclusions that would otherwise remain hidden in purely correlational analyses (Pearl, 2009).

Central to this view is the idea that every model encodes, implicitly or explicitly, a set of assumptions about the data-generating process that produced the observations under study. The richer and more explicit these assumptions are about the underlying process, the more information we are able to extract from the data itself. In other words, the assumptions a model makes are not a liability to be minimized but a resource to be used purposefully: they determine how much of the structure latent in the data-generating process can be recovered, and thus how much genuine insight a cognitive model of language can ultimately offer.

One example of latent-parameter inference through a well-reasoned model, which highlights the usefulness for probabilistic pragmatic models for the interpretation of experimental data, is *latent semantic inference*. The vague quantifier *many* can convey quite different absolute numbers in sentences like in (3), arguably because its denotation is sensitive to general expectations about the relevant quantities, such as the number of shoes owned, number of pages read.

(1) Jake owns many pairs of shoes.

(2) Joe read many pages in his book last week.

Fernando and Kamp (1996) hypothesized that, despite this context-dependence, there is still a stable semantic core meaning based on an absolute threshold on the cumulative distribution of the expectation about the relevant quantity. Without going into detail of this particular case study (Schöller and Franke, 2017a), the general recipe for latent semantic inference through a pragmatics model is that we assume that there is a latent parameter θ, which determines the semantic meaning of an expression, which in turn influences the speaker's inclination to respond in particular ways to tasks posed during a suitable experiment. [MF: insert picture] So, the speaker's behavior $\pi_S(a \mid \theta)$ in the experimental task should be conceived of as a function of that hidden semantic parameter. From the point of the view of the experimenter, we can then take the empirically observed data, use Bayes rule much like the pragmatic listener model does, and backwards-infer likely values of θ based on the experimental data. The more independent justification there is—from theoretical considerations or other experimental work—to assume that the functional form of the stipulated policy is (roughly) correct, the more we can use the model as a lens to learn about theoretically meaningful parameters.

This is the power that probabilistic pragmatic modeling can unfold if used as part of the data-analytical pipeline: a theoretically informed model can become the statistical model itself. Multiple models, differing in meaningful theoretical distinctions can then be stringently contrasted through statistical model comparison. Using this approach, van Tiel, Franke, and Sauerland (2021) compared models of quantifier use to distinguish between a standard logical semantics based on thresholds [MF: ref] and a semantics in terms of distance to a prototype [MF: refs], using a probabilistic pragmatic speaker policy based on informativity-sensitive utilities as a model of the data-generating process with latent parameters inferred for either thresholds or prototypes of quantifiers. Similar methods have been used in the domain of quantifiers (Carcassi and Szymanik, 2021; van Tiel, Franke, and Sauerland, 2022; Tiel et al., 2026), vague gradable adjectives Cremers (2022) uses a complex RSA-style model for Bayesian parameter inference for vague gradable adjectives, generics Tessler and Goodman (2019), and general (syntactic) ambiguity (Franke and Bergen, 2020), different models of referential reasoning (Qing and Franke, 2015; Franke and Degen, 2016), or intensifiers (Schöller and Franke, 2017b; Zhao, 2019). [MF: more and better refs? Anton Benz, Zhao?]

- problem of link functions

4 Towards a computational cognitive science of communication

The blueprint for probabilistic pragmatics from Section §2 was, for the listener, Bayes rule to reverse-engineer the likely mental states that caused linguistic behavior, and, for the speaker, soft-max policy choice to optimize heuristics for "good utterances in context." This suggests that linguistic communication is a pair of, so to speak, mind-*reconstruction* (listener) and mind-*anticipation* (speaker), which is —ideally— jointly approximately optimal. This joint-optimization picture has inspired analyses in terms of game-theoretic solutions, e.g., variations on Nash equilibria (e.g., Parikh, 1988) or higher-order iterations of best responses (e.g. Pavan, 2013; Rothschild, 2013; Franke and Jäger, 2014). However, while these approaches can tell us which behavioral policies are jointly optimal, they do not give a processing-level account of how human cognition implements (or finds) these strategies. This criticism is reminiscent of the motivation behind *relevance theory* (Sperber and Wilson, 1995; Wilson and Sperber, 2004), namely to embed a theory of pragmatic language use in the broader picture of human cognition and its evolution. Yet while I consider what relevance theory wanted to do (see pragmatics in a broader cognitive picture) to be right, I also side with those who see flaws in relevance theory's realization of its ambitions (e.g., Gazdar and Good, 1982; Levinson, 1989). However, using explicit probabilistic models as theory-informed statistical models, it is possible to develop a computational cognitive science of human linguistic computation, especially when integrating recent insights and tools from cognitive neuroscience and machine learning.

Indeed, I argue that both game-theoretic and probabilistic pragmatics in the tradition of the RSA framework are misspecified as reasonable computational cognitive models of human pragmatic communication because they start of from the wrong problem description, i.e., from a wrong model of what context-dependent linguistic communication requires human interlocutors to do, given the general make-up of human information processing. This, too, is parallel to early arguments proposed by relevance theory, namely that linguistic communication is not an information-theoretic encoding and decoding problem, but a genuinely *inferential process*. However, in my view, the main difference between information-theoretic encoder-decoder architectures and human linguistic communication is not that the latter is inferential, while the former is not (information-theory also standardly uses Bayesian inference as the decoding strategy, just like probabilistic pragmatics does). Rather it is because human communication is based on inference in a massively uncertain, richly structured space, including other minds and their dynamics (as influenced by speech and other actions and events), and uncertainty about what knowledge to assume for context-enriched efficient information flow. Standard game-theoretic models like Lewisian signaling games (Lewis, 1969) or probabilistic models in the RSA framework also essentially only provide solutions for an encoding-decoding problem when they assume fixed sets of interpretation options, linguistic actions and possible reactions to them. This is because they do not factor in the problem that human cognition, outside of stylized laboratory experiments, must operate in an open-ended hypothesis space, which is a problem that language production itself must be catered to mitigate: if efficient language production is "mind-anticipation" then this must entail anticipating "mind-dynamics" in an open-ended world. In other words, future models of pragmatic reasoning should ideally stop being agnostic about the role of core cognitive processes for reasoning about shared or recruitable context information, such as awareness or memory. Instead, what is required is an explicit cognitive turn in modeling that integrates probabilistic reasoning about what the relevant context is, including the perspectives of other agents. [MF: jump to 'context' is too abrupt; maybe motivate 'awareness' and 'meta-cognition' more swiftly?]

To illustrate the importance of reasoning about awareness and attention, consider an example of written communication as in the following stylized exchange:

(3) Sender: If I were a ~~German~~ person who didn't like spicy food, I'd not go there.
Receiver (a German): Touché. But also: ouch!

This is an example of a general and quite productive class of pragmatic inferences, which I here call *strike-through implicature* [MF: TODO: check if literature on this phenomenon exists; maybe relevance theory on inferences from pretend errors]. The writer chose to write a passage and mark it as deleted or to be ignored, in full awareness that the reader will recognize that the speaker has thereby made the reader aware of the speaker's as-if initial consideration of the elided text. If we consider the standard picture of pragmatic reasoning about alternative utterances the speaker could have made, we could analyze the case as follows: an utterance of (3) with strike-through may signal that the speaker has considered both of the alternatives (4) below, ended up preferring (4b) but also wants the recipient to know that they have deliberately NOT chosen the alternative (4a).

(4) a. If I were a German person who didn't like spicy food, I'd not go there.
b. If I were a person who didn't like spicy food, I'd not go there.

For current argumentative purposes, the important observation is that *if* the speaker had only written (**??**), the recipient would never even have considered that the speaker deliberately did NOT chose (**??**). Much more could be said about strike-through implicatures and like-minded cases of pretend-errors in spoken language (think: "Whoops!? Did I really just say that?"), but the key take-away for the present argument is, quite general and imprecisely stated, that pragmatic inferences draw on the complex, counterfactual dynamics of awareness and attention that speakers can anticipate and listeners can recover. Some work in formal semantics and pragmatics has explored effects of awareness or attention dynamics [MF: REFs] and while there are logical [MF: REFs] and game-theoretic [MF: REFs] frameworks for modeling reasoning about unawareness, a systematic treatment from the point of view of a cognitive model of attention dynamics in pragmatic communication is missing.

Here is a quite different example that serves as an additional motivation of the central inclusion of dynamic perspectives in pragmatic reasoning, but also motivates the importance of considering single-agent meta-cognitive processes that may steer the awareness dynamics. Cummins and Franke (2021) discuss the argumentative use of quantifying expressions based on selected cases of UK universities self-reporting on the results of the 2014 Research Excellence Framework, an example of which is (5)

(5) Royal Holloway is within the top 25 per cent of United Kingdom universities for research rated 'world-leading' or 'internationally excellent'.

When reading this statement, multiple layers of depth of pragmatic interpretation may be applied, where interpretative depth does not amount to iterated social reasoning, but the degree to which the wider strategic context of utterance production is taken into account. On a first, extremely superficial reading, the statement could be taken as just a random informative remark. With slightly more pragmatic awareness, so to speak, a reader may realize that this statement is likely intended as some kind of argument for *something* in the sense that it provides suggestive evidence in favor of a particular view. It may then not take much to fill this *something* with a contextually salient explanation of *what* the statement is likely arguing for (in the case at hand, this would likely be that Royal Holloway is a good university, according to the REF 2014). So far, these layers may come easy and automatic, but with even more depth, a recipient may realize that the argument made is likely not particularly strong (e.g., because the top 25% ranking is reported, apparently, only for a cherry-picked subset of criteria reported on in the REF 2014). Once more, it does not seem to be the case that these layered depths of pragmatic reasoning correspond to increasingly deeper nestings of iterate Theory-of-Mind like reasoning. Rather, they seem to relate to how much of the utterance-generating process the interpreter cares to or is able to attend to, which directly relates to notions like automatic vs. controlled processes and executive functions in cognitive psychology. [MF: why just namedrop these here? spell out?]

> - **TODO** transition into meta-cognition & model revision. What could trigger an agent to 'reason deeper'? Attention allocation based on meta-cognitive confidence → model revision triggered by accumulated intrinsic error signals.
>
> - **TODO** optimal-rational analysis → which problem is solved with which ingredients? But acknowledge that the ingredients (cognitive make-up) define part of the problem and thus must be acknowledged in both problem+solution.
>
> - **TODO** überleitung. The picture emerging from these cases: communicators may be variously aware or unaware of more or less possibly relevant features of the context (including alternative interpretations and utterances, strategic incentives and goals). Common speech-practices are clearly aware of these attentional and awareness dynamics: people forget, so we have practices of reminding and even joint reminiscing; people are inattentive, so we know that simple reminders work, e.g., we may accept "Thanks." as the right reaction to a question like "Did you call Jake?"; and at a finer-grained temporal resolution, we may find how referential descriptions may be adapted during the search process to the shifting visual access of the searching listener [MF: REFs]; etc. ... For that reason, I argue that a future computational cognitive science of communication (CCSC) needs to consider modeling that includes (at least basic principles) of the dynamics of attention, meta-cognition, and possibly the online temporal dynamics of language production and interpretation.

References

Asimov, Isaac (1988). *The Relativity of Wrong*. Doubleday.

Austin, John L. (1962). *How to do Things with Words: The William James Lectures delivered at Havard University in 1955*. Oxford: Clarendon.

Benz, Anton (2006). "Utility and Relevance of Answers". In: *Game Theory and Pragmatics*. Ed. by Anton Benz, Gerhard Jäger, and Robert van Rooij. Hampshire: Palgrave, pp. 195–219.

Benz, Anton and Robert van Rooij (2007). "Optimal Assertions and what they Implicate". In: *Topoi* 26, pp. 63–78.

Bergen, Leon, Roger Levy, and Noah D. Goodman (2012). "That's what she (could have) said: How alternative utterances affect language use". In: *Proceedings of the 34th Annual Meeting of the Cognitive Science Society*. Vol. 34, pp. 120–125.

— (2016). "Pragmatic Reasoning through Semantic Inference". In: *Semantics & Pragmatics* 9.20, pp. 1–83.

Blutner, Reinhard and Henk Zeevat, eds. (2004). *Optimality Theory and Pragmatics*. Palgrave MacMillan.

Box, George E. P. (1979). "Robustness in the strategy of scientific model building". In: *Robustness in Statistics*. Ed. by R. L. Launer and G. N. Wilkinson. Cambridge, MA: Academic Press, pp. 201–236.

Bunt, Harry and William Black (2000). *Abduction, Belief and Context in Dialogue*. Amsterdam, Philadelphia: John Benjamins.

Burnett, Heather (2019). "Signalling games, sociolinguistic variation and the construction of style". In: *Linguistics and Philosophy* 42, pp. 419–450.

Carcassi, Fausto and Michael Franke (2023). "How to handle the truth: A model of politeness as strategic truth-stretching". In: *Proceedings of CogSci*. Ed. by M. Goldwater et al., pp. 222–228.

Carcassi, Fausto and Jakub Szymanik (2021). "An alternatives account of 'most' and 'more than half'". In: *Glossa: a journal of general linguistics* 6.1.

Carcassi, Fausto, Hening Wang, et al. (2026). "What guides utterance choice in argumentative language use?" In: *CogSci 48*. Ed. by G. Boleda et al., pp. 389–395.

Cremers, Alexandre (2022). "A Rational Speech-Act model for the pragmaticuse of vague terms in natural language". In: *Proceedings of the 44th Annual Meeting of the Cognitive Science Society*. Ed. by J. Culbertson et al. Vol. 44, pp. 149–155.

Cummins, Chris and Michael Franke (2021). "Rational Interpretation of Numerical Quantity in Argumentative Contexts". In: *Frontiers in Communication* 6, p. 89.

Cummins, Chris and Napoleon Katsos, eds. (2019). *The Oxford Handbook of Experimental Semantics and Pragmatics*. Oxford: Oxford University Press.

Degen, Judith (2023). "The Rational Speech Act Framework". In: *Annual Review of Linguistics* 9.1, pp. 519–540.

Degen, Judith and Michael Franke (2012). "Optimal Reasoning About Referential Expressions". In: *Proceedings of SemDial 2012 (SeineDial): The 16th Workshop on the Semantics and Pragmatics of Dialogue*. Ed. by Sarah Brown-Schmidt, Jonathan Ginzburg, and Staffan Larsson, pp. 2–11.

Degen, Judith, Michael Franke, and Gerhard Jäger (2013). "Cost-Based Pragmatic Inference about Referential Expressions". In: *Proceedings of the 35th Annual Meeting of the Cognitive Science Society*. Ed. by Markus Knauff et al. Austin, TX: Cognitive Science Society, pp. 376–381.

Fernando, Tim and Hans Kamp (1996). “Expecting Many”. In: *Proceedings of SALT 6*. Ed. by Teresa Galloway and Justin Spence. Ithaca, NY: Cornell University, pp. 53–68.

Frank, Michael C., Andrés Gómez Emilsson, et al. (2016). “Rational speech act models of pragmatic reasoning in reference games”.

Frank, Michael C. and Noah D. Goodman (2012). “Predicting Pragmatic Reasoning in Language Games”. In: *Science* 336.6084, p. 998.

Franke, Michael and Leon Bergen (2020). “Theory-driven statistical modeling for semantics & pragmatics: A case study on grammatically generated implicature readings”. In: *Language* 96.2, e77–e96.

Franke, Michael and Judith Degen (2016). “Reasoning in Reference Games: Individual- vs. Population-Level Probabilistic Modeling”. In: *PLoS ONE* 11.5, e0154854.

— (2023). “The softmax function: Properties, motivation, and interpretation”. In.

Franke, Michael and Gerhard Jäger (2014). “Pragmatic Back-and-Forth Reasoning”. In: *Semantics, Pragmatics and the Case of Scalar Implicatures*. Ed. by Salvatore Pistoia Reda. Palgrave Studies in Pragmatics Language and Cognition. New York: Palgrave MacMillan. Chap. 7, pp. 170–200.

— (2016). “Probabilistic pragmatics, or why Bayes’ rule is probably important for pragmatics”. In: *Zeitschrift für Sprachwissenschaft* 35.1, pp. 3–44.

Gazdar, Gerald (1979). *Pragmatics: Implicature, Presupposition, and Logical Form*. New York: Academic Press.

Gazdar, Gerald and David Good (1982). “On a notion of relevance”. In: *Mutual Knowledge*. Ed. by N. Smith. Academic Press London, pp. 88–100.

Goodman, Noah D. and Daniel Lassiter (2014). “Probabilistic Semantics and Pragmatics: Uncertainty in Language and Thought”. Chapter draft, Stanford University.

Goodman, Noah D. and Andreas Stuhlmüller (2013). “Knowledge and Implicature: Modeling Lanuage Understanding as Social Cognition”. In: *Topics in Cognitive Science* 5, pp. 173–184.

Grice, Paul Herbert (1975). “Logic and Conversation”. In: *Syntax and Semantics, Vol. 3, Speech Acts*. Ed. by Peter Cole and Jerry L. Morgan. New York: Academic Press, pp. 41–58.

Hobbs, Jerry R., Mark Stickel, and Paul Martin (1993). “Interpretation as Abduction”. In: *Artificial Intelligence* 63, pp. 69–142.

Kadmon, Nirit (2001). *Formal Pragmatics*. Oxford: Wiley-Blackwell.

Kao, Justine T. et al. (2014). “Nonliteral Understanding of Number Words”. In: *PNAS* 111.33, pp. 12002–12007.

Karttunen, Lauri (1973). “Presuppositions of Complex Sentences”. In: *Linguistic Inquiry* 4, pp. 167–193.

Lai, Lucy and Samuel J. Gershman (Apr. 2024). “Human decision making balances reward maximization and policy compression”. In: *PLOS Computational Biology* 20.4. Ed. by Thomas Serre, e1012057. ISSN: 1553-7358.

Levinson, Stephen C. (1989). “A Review of Relevance”. In: *Journal of Linguistics* 25.2, pp. 455–472.

Lewis, David (1969). *Convention. A Philosophical Study*. Cambridge, MA: Harvard University Press.

Lieder, Falk and Thomas L. Griffiths (2019). “Resource-rational analysis: Understanding human cognition as the optimal use of limited computational resources”. In: *The Behavioral and Brain Sciences* 43, e1.

Meibauer, Jörg and Markus Steinbach, eds. (2011). *Experimental Pragmatics/Semantics*. Amsterdam, Philadelphia: John Benjamins.

Merin, Arthur (1994). “Algebra of Elementary Social Acts”. In: *Foundations of Speech Act Theory. Philosophical and Linguistic Perspectives*. Ed. by Savas L. Tsohatzidis. London and New York: Routledge, pp. 234–263.

— (1999). “Information, Relevance, and Social Decisionmaking: Some Principles and Results of Decision-Theoretic Semantics”. In: *Logic, Language and Computation*. Ed. by Lawrence C. Moss, Jonathan Ginzburg, and Maarten de Rijke. Vol. 2. CSLI Publications, pp. 179–221.

Noveck, Ira A. (2018). *Experimental Pragmatics: The Making of a Cognitive Science*. New York: Cambridge University Press.

Noveck, Ira A. and Dan Sperber, eds. (2004). *Experimental Pragmatics*. Hampshire: Palgrave MacMillan.

Nowak, Martin A. and David C. Krakauer (1999). “The Evolution of Language”. In: *PNAS* 96, pp. 8028–8033.

Parikh, Prashant (1988). “Language and strategic inference”. PhD thesis. Stanford University.

Pavan, Sascia (2013). “Scalar Implicatures and Iterated Admissibility”. In: *Linguistics and Philosophy* 36, pp. 261–290.

Pearl, Judea (Jan. 2009). “Causal inference in statistics: An overview”. In: *Statistics Surveys* 3.none. ISSN: 1935-7516.

Qing, Ciyang and Michael Franke (2015). “Variations on a Bayesian Theme: Comparing Bayesian Models of Referential Reasoning”. In: *Bayesian Natural Language Semantics and Pragmatics*. Ed. by Henk Zeevat and Hans-Christian Schmitz. Language, Cognition and Mind. Berlin: Springer, pp. 201–220.

Rothschild, Daniel (2013). “Game Theory and Scalar Implicatures”. In: *Philosophical Perspectives* 27.1, pp. 438–478.

Russell, Benjamin (2012). “Probabilistic Reasoning and the Computation of Scalar Implicatures”. PhD thesis. Brown University.

Schöller, Anthea and Michael Franke (2017a). “Semantic values as latent parameters: Testing a fixed threshold hypothesis for cardinal readings of *few* & *many*”. In: *Linguistic Vanguard* 3.1.

— (2017b). “*Surprisingly*: Marker of Surpirse readings or Intensifier?” In: *Proceedings of CogSci 39*. Ed. by Glenn Gunzelmann et al. Austin, TX: Cognitive Science Society, pp. 3070–3075.

Scontras, Gregory, Michael Henry Tessler, and Michael Franke (2021). *A practical introduction to the Rational Speech Act modeling framework*.

Searle, John R. (1969). *Speech Acts: An Essay in the Philosophy of Language*. Cambridge University Press.

Skyrms, Brian (2010). *Signals: Evolution, Learning, and Information*. Oxford: Oxford University Press.

Sperber, Dan and Deirdre Wilson (1995). *Relevance: Communication and Cognition (2nd ed.)* Oxford: Blackwell.

Sprouse, Jon (2007). “A program for experimental syntax: Finding the relationship between acceptability and grammatical knowledge”. PhD thesis. University of Maryland.

— ed. (2023). *The Oxford Handbook of Experimental Syntax*. Oxford University Press.

Stalnaker, Robert (1973). “Presupposition”. In: *The Journal of Philosophical Logic* 2, pp. 447–457.

Tessler, Michael Henry and Noah D. Goodman (2019). “The Language of Generalization”. In: *Psychological Science* 126.3, pp. 395–436.

Tiel, Bob van et al. (2026). “Bespoke Probabilistic Modelling Reveals Subtle Effects of Autism on Pragmatic Optimisation in the Expression of Quantification”. In: *Computational Brain & Behavior*.

van Tiel, Bob, Michael Franke, and Uli Sauerland (2021). “Probabilistic pragmatics explains gradience and focality in natural language quantification”. In: *Proceedings of the National Academy of Sciences* 118.

— (2022). “Meaning and use in the expression of estimative probability”. In: *Open Mind: Discoveries in Cognitive Science* 6, pp. 250–263.

Wilson, Deirdre and Dan Sperber (2004). “Relevance Theory”. In: *Handbook of Pragmatics*. Ed. by Laurence R. Horn and Gregory Ward. Oxford: Blackwell, pp. 607–632.

Yoon, Erica J. et al. (2020). “Polite Speech Emerges From Competing Social Goals”. In: *Open Mind: Discoveries in Cognitive Science* 4, pp. 71–87.

Zeevat, Henk (2014). *Language Production and Interpretation: Linguistics meets Cognition*. Current Research in the Semantics/Pragmatics Interface. Leiden: Brill.

Zhao, Zhuoye (2019). “Interpreting Intensifiers for Relative Adjectives: Comparing Models and Theories”. In: *At the Intersection of Language, Logic, and Information*. Springer Berlin Heidelberg, pp. 213–224. ISBN: 9783662596203.